

MED 610 Examination

Clinical Reasoning and Differential Diagnosis – Answer Key

1. **(B)** Aortic dissection.

Classic presentation: sudden, severe, tearing chest pain radiating to the back, with blood pressure differential between arms (indicating involvement of subclavian artery). This is a medical emergency requiring immediate imaging (CT angiography or transesophageal echo) and management.

2. **(B)** Diabetic ketoacidosis.

Young patient with hyperglycemia, polyuria, polydipsia, weight loss, and ketonuria suggests type 1 diabetes presenting with DKA. The presence of ketones distinguishes this from simple hyperglycemia or hyperosmolar state (which typically occurs in type 2 diabetes).

3. **(C)** Obstructive jaundice (posthepatic).

Elevated conjugated (direct) bilirubin with elevated alkaline phosphatase indicates biliary obstruction. Dark urine (conjugated bilirubin excretion) and pale stools (absent stercobilin) confirm obstruction. Causes include gallstones, pancreatic cancer, or biliary strictures.

4. **(B)** Squamous cell carcinoma.

Central location with hilar involvement is characteristic of squamous cell carcinoma, which arises from central bronchi. It's strongly associated with smoking. Small cell is also central but more aggressive with early metastasis. Adenocarcinoma is typically peripheral.

5. *Sample answer:*

Systematic approach to acute abdominal pain:

1. Anatomical consideration (by quadrant):

- **RUQ:** Cholecystitis, hepatitis, liver abscess
- **RLQ:** Appendicitis, ectopic pregnancy, ovarian torsion, Crohn's disease
- **LUQ:** Splenic rupture, gastric ulcer, pancreatitis
- **LLQ:** Diverticulitis, ectopic pregnancy, ovarian torsion
- **Epigastric:** Peptic ulcer, pancreatitis, MI
- **Diffuse:** Peritonitis, mesenteric ischemia, bowel obstruction

2. Character of pain:

- Colicky (intermittent): Obstruction (kidney stone, bowel obstruction)
- Constant, severe: Perforation, ischemia, severe inflammation
- Burning: Peptic ulcer disease, GERD

3. Associated symptoms:

- Fever: Infection (appendicitis, diverticulitis, cholecystitis)
- Vomiting: Obstruction, acute abdomen
- Diarrhea: Gastroenteritis, inflammatory bowel disease
- Hematemesis/melena: GI bleeding

4. Initial tests:

- Labs: CBC, CMP, lipase, liver enzymes, pregnancy test (females), urinalysis
- Imaging: Upright chest X-ray (free air), abdominal X-ray, ultrasound, CT scan

6. *Sample answer:*

Differential diagnosis of heart failure:

Left-sided heart failure:

- **Causes:** Ischemic heart disease, hypertension, aortic/mitral valve disease, cardiomyopathy
- **Features:** Dyspnea, orthopnea, paroxysmal nocturnal dyspnea (PND), pulmonary edema (rales on auscultation), S3 gallop
- **Mechanism:** Inability to pump blood forward → pulmonary congestion

Right-sided heart failure:

- **Causes:** Left heart failure (most common), pulmonary hypertension, tricuspid/pulmonary valve disease, cor pulmonale
- **Features:** Jugular venous distension (JVD), hepatomegaly, ascites, peripheral edema, hepatojugular reflux
- **Mechanism:** Inability to pump blood forward from right ventricle → systemic venous congestion

Biventricular failure:

- Combination of left and right heart failure features
- Most common presentation in advanced heart failure

This patient: Has features of both left (dyspnea, orthopnea) and right (JVD, edema) heart failure, indicating biventricular failure.

7. *Sample answer:*

Fever of Unknown Origin (FUO): Fever $> 38.3^{\circ}\text{C}$ for more than 3 weeks without diagnosis despite initial evaluation.

Major categories:

- (a) **Infections** (30–40%): Tuberculosis, endocarditis, abscesses, HIV/opportunistic infections
- (b) **Malignancies** (20–30%): Lymphoma, leukemia, renal cell carcinoma
- (c) **Autoimmune/inflammatory** (10–20%): Still's disease, SLE, vasculitis, inflammatory bowel disease
- (d) **Miscellaneous:** Drug fever, factitious fever, pulmonary embolism

Initial workup:

- **History:** Travel, exposures, medications, risk factors

- **Exam:** Careful physical including lymph nodes, skin, heart murmurs, hepatosplenomegaly
- **Labs:** CBC with differential, ESR/CRP, blood cultures (multiple sets), urinalysis, LFTs, ANA, RF
- **Imaging:** Chest X-ray, CT chest/abdomen/pelvis (look for lymphadenopathy, masses, abscesses)
- **Special tests:** PPD/QuantiFERON (TB), HIV test, echocardiogram (endocarditis), lumbar puncture if indicated

Consider empiric antibiotics only if patient is hemodynamically unstable.

8. *Sample answer:*

Differential diagnosis of altered mental status in elderly:

Key distinction - Delirium vs. Dementia:

- **Delirium:** Acute onset (hours to days), fluctuating course, inattention prominent, reversible with treatment of underlying cause
- **Dementia:** Chronic onset (months to years), progressive, insidious, memory impairment prominent, usually irreversible

Acute medical causes to urgently rule out:

- (a) **Metabolic:** Hypoglycemia, hyponatremia, hypernatremia, hypercalcemia, hepatic encephalopathy, uremia
- (b) **Infectious:** UTI, pneumonia, sepsis, meningitis/encephalitis
- (c) **Cardiovascular:** Stroke, MI, heart failure, arrhythmias causing hypoperfusion
- (d) **Respiratory:** Hypoxia from any cause
- (e) **Medications:** Anticholinergics, benzodiazepines, opioids, polypharmacy
- (f) **Withdrawal:** Alcohol, benzodiazepines
- (g) **Neurological:** Seizures (postictal state), subdural hematoma, increased ICP
- (h) **Endocrine:** Thyroid dysfunction (myxedema coma, thyroid storm), Addisonian crisis

Approach:

- ABCDE assessment, vital signs
- Finger-stick glucose immediately
- Labs: CBC, CMP, calcium, TSH, urinalysis, blood cultures
- CT head (rule out stroke, hemorrhage, mass)
- Consider lumbar puncture if meningitis suspected

Remember: Delirium often superimposed on dementia in elderly. Treat reversible causes aggressively.